School of Science, Computing and Emerging Technologies

Applied Machine Learning

Lab 5. Transfer Learning

Overview

This lab focuses on a multi-class image classification task. You are required to apply transfer learning techniques to solve the problem.

Dataset

We will use the Kaggle dataset titled "**Deep Learning Practice - Image Classification**":
https://drive.google.com/drive/folders/1dwGzQwooU4PT642faBxBj_CS6OB_FiFE

The dataset contains two folders:

- train: 10,000 images across 10 classes (1,000 images per class)
- test: unlabeled images (not used in this lab)

For this lab, only the train folder is used. You must split it into training and validation sets (eg. 80/20 split).

Model

Build an image classification model using a pre-trained network, implementing two approaches:

- Feature Extraction: Freeze the base model, and train only the newly added classification layers
- Fine-Tuning: Unfreeze some of the top layers of the base model, and continue training the model

Compare both methods in terms of accuracy and training time.

Report

Create a short report (1 page) including:

- The pre-trained model used
- Your model architecture diagram
- A comparison of feature extraction vs fine-tuning results

Submission

Submit your source code and report via Canvas and demonstrate your work to the tutor during the lab session.